

Fundamentals of Artificial Intelligence

Georgetown Pre-Advanced eDiscovery Institute Webinar

July 18, 2023

Professor Maura R. Grossman


20th ANNUAL
GEORGETOWN LAW
Advanced eDiscovery Institute

Welcome

20th Annual Georgetown Law AEDI

- Hold the date: November 9 - 10
- This year's Keynote Speakers

Dates of Additional Pre-Conference Webinars

Your Speaker Today…

- Dr. Maura R. Grossman
- Materials have been sent to you in an email

Agenda for Today

- What Artificial Intelligence Is and Why Now?
- How AI Works
- A Quick Tutorial on Generative AI
- What AI Can Do
- Some Issues Related to AI Including Job Displacement





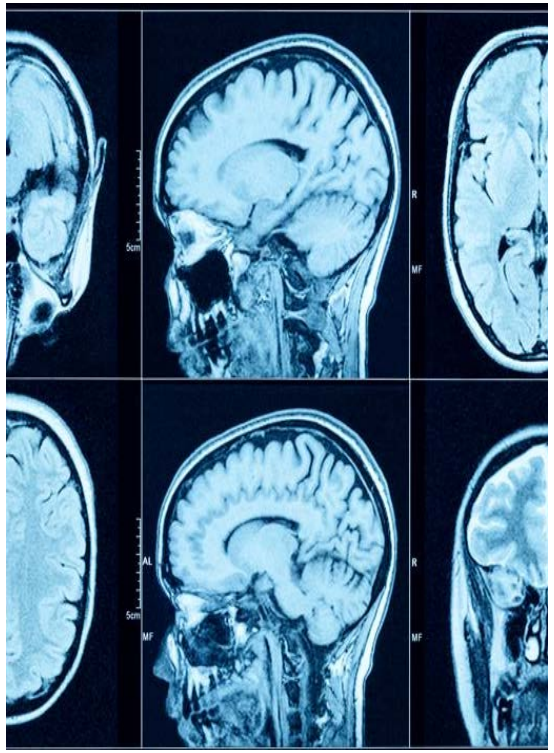
What AI Is and Why Now

WHAT IS "ARTIFICIAL INTELLIGENCE"?

- Umbrella term first used at Dartmouth in 1956
- Computers doing intelligent things (*i.e.*, performing cognitive tasks) once thought to be the sole province of humans
- Not a single technology or function
- Whatever computers can't do … until they can
- Called "software" after we get used to it
- Different from automation and robotics
- Generally involves algorithms, machine learning, and/or natural language processing ("NLP")

TYPES OF AI

Narrow (Weak) AI vs. General (Strong) AI

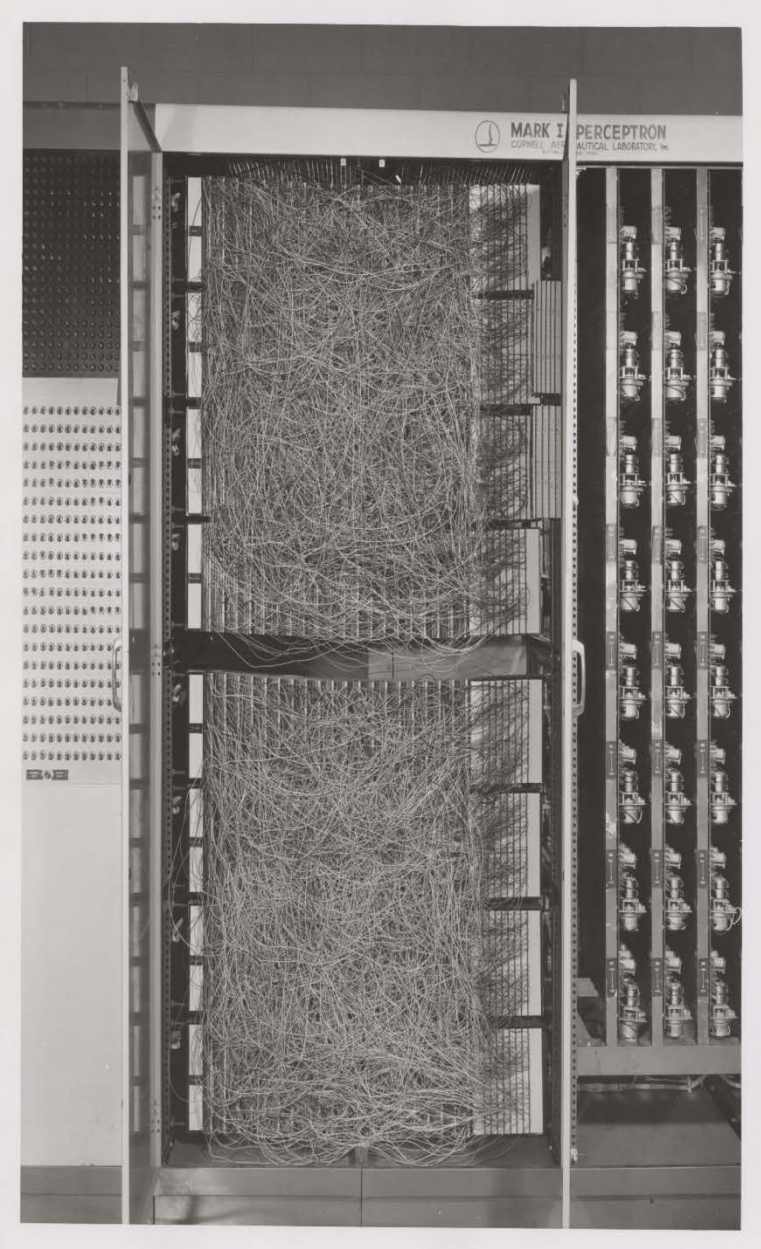




BUT ISN'T MACHINE LEARNING AN EMERGING TECHNOLOGY?



- Logistic Regression – 1944
- Stochastic Gradient Descent ("SGD") – 1950s
- Artificial Neural Network ("ANN") – 1958
- Hidden Markov Model – late 1950s/1960
- Decision Tree – 1975
- Expectation Maximization – 1977
- Q-Learning – 1989
- Support Vector Machine ("SVM") – 1990s
- K-Nearest Neighbor ("K-NN") – 1992
- Random Forest – 2001



MORE COMPUTING POWER

Rosenblatt's Mark I Perceptron
Cornell University, 1957

MORE DATA

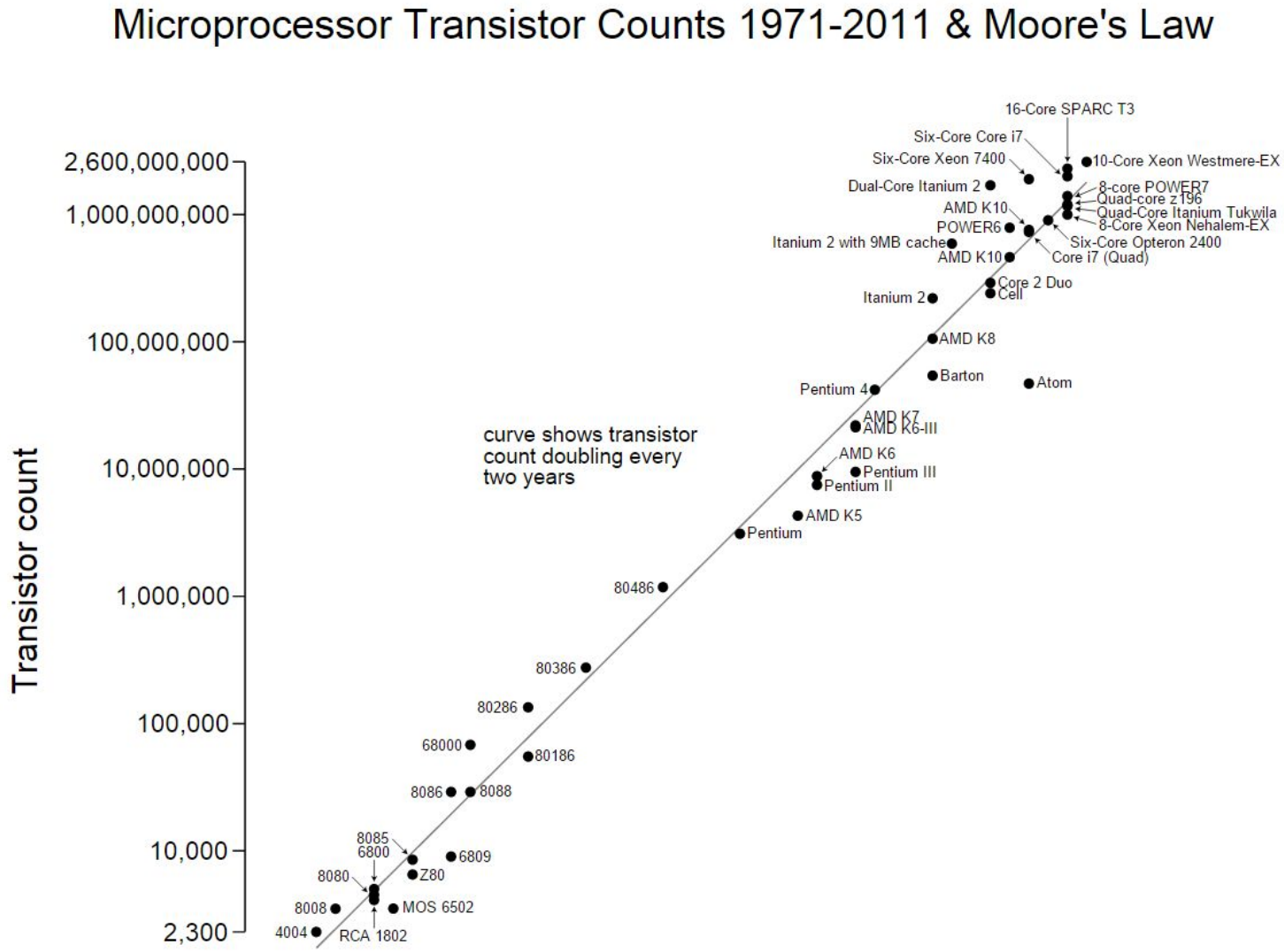


KRYDER'S LAW

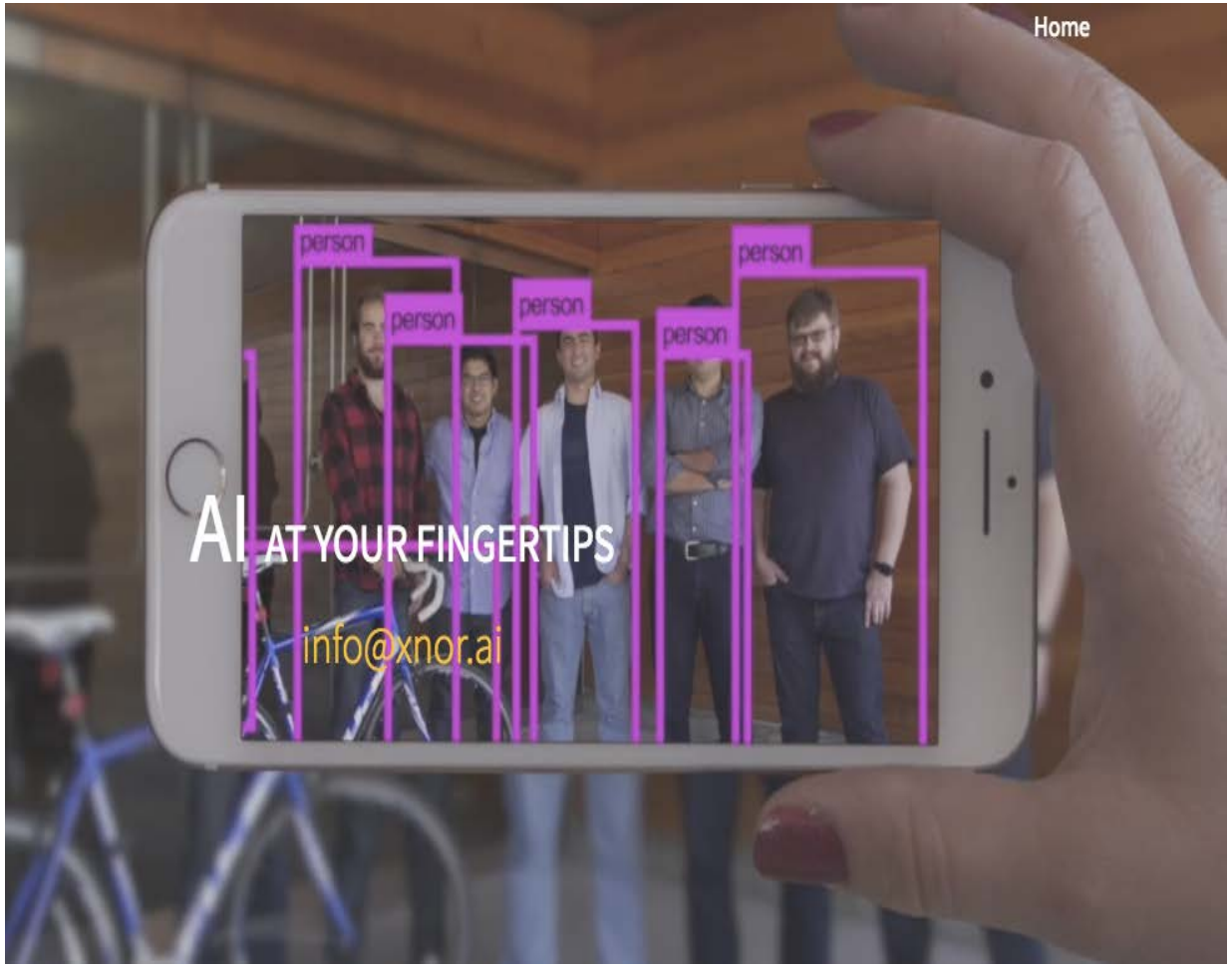
- **Purchase price for storage of one GB of data over 20 years:**
  - **1981 – $300,000**
  - **1987 – $50,000**
  - **1990 – $10,000**
  - **1994 – $1,000**
  - **1997 – $100**
  - **2000 – $10**
  - **2004 – $1**
  - **2011 – $0.10**
  - **2021 – Practically free**

MOORE’S LAW

The number of transistors on a microchip (and hence the speed of processing) doubles every 18 to 24 months



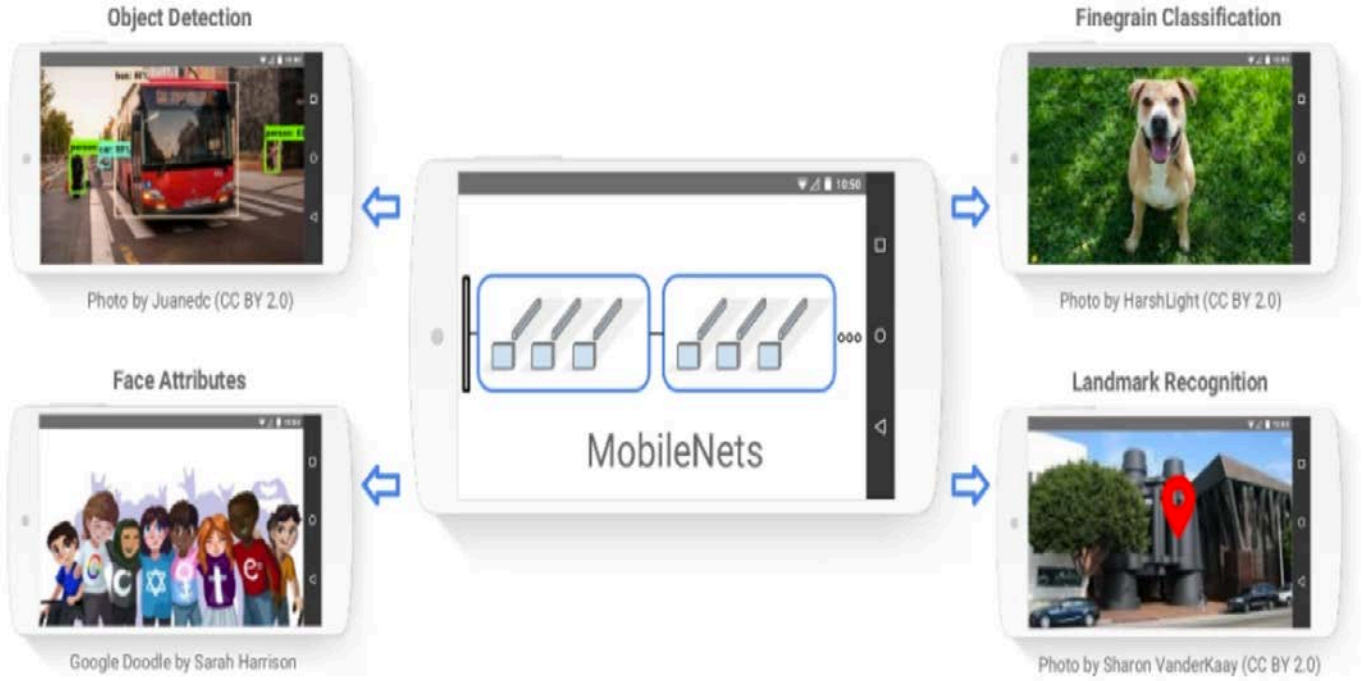
NEXT GENERATION HARDWARE



OPEN SOURCE COMMUNITIES LOWER BARRIERS TO ENTRY

Google open-sources mobile-first computer vision models for TensorFlow

BLAIR HANLEY FRANK @BELRIL JUNE 14, 2017 11:09 AM

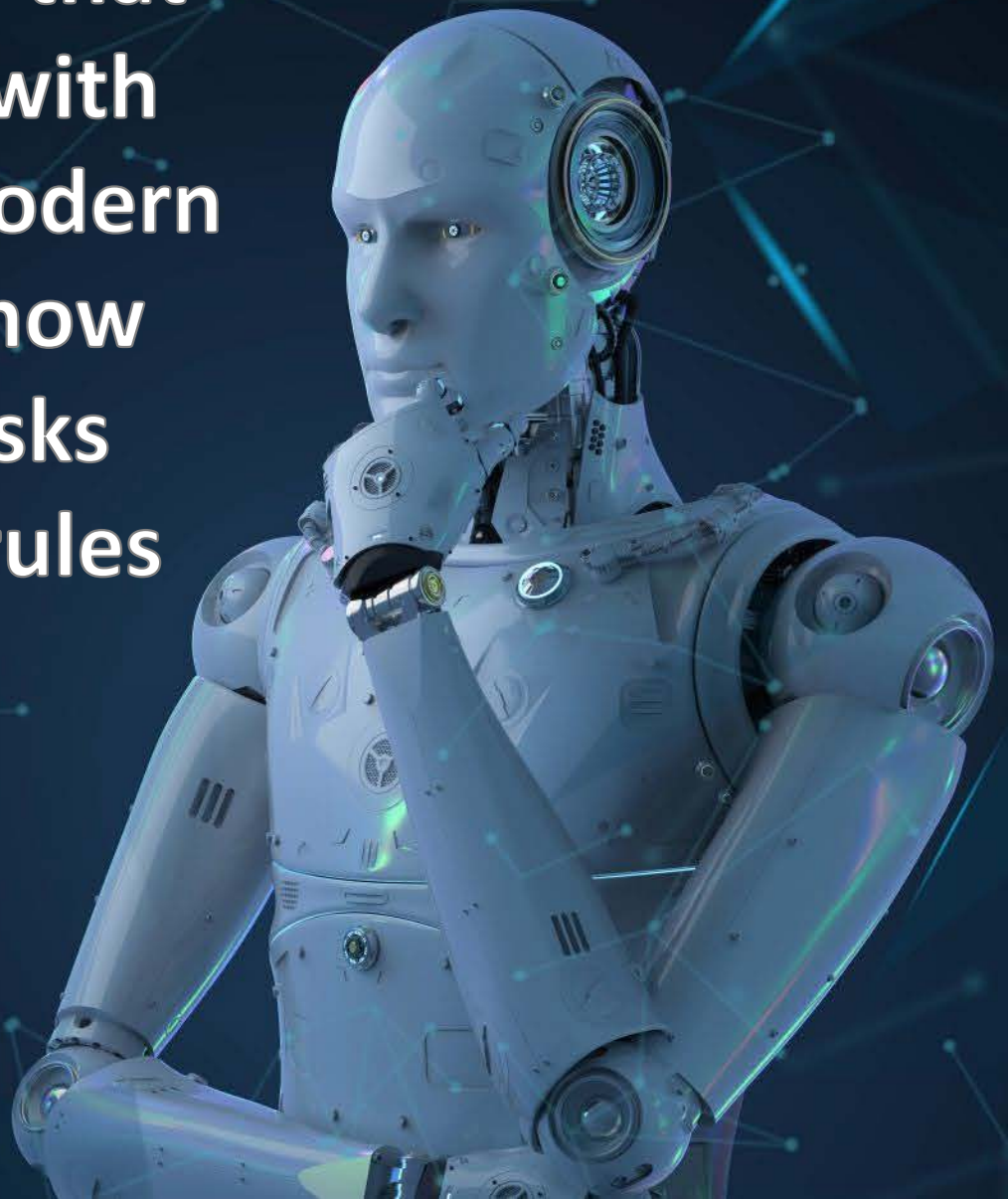


FOLLOW THE MONEY

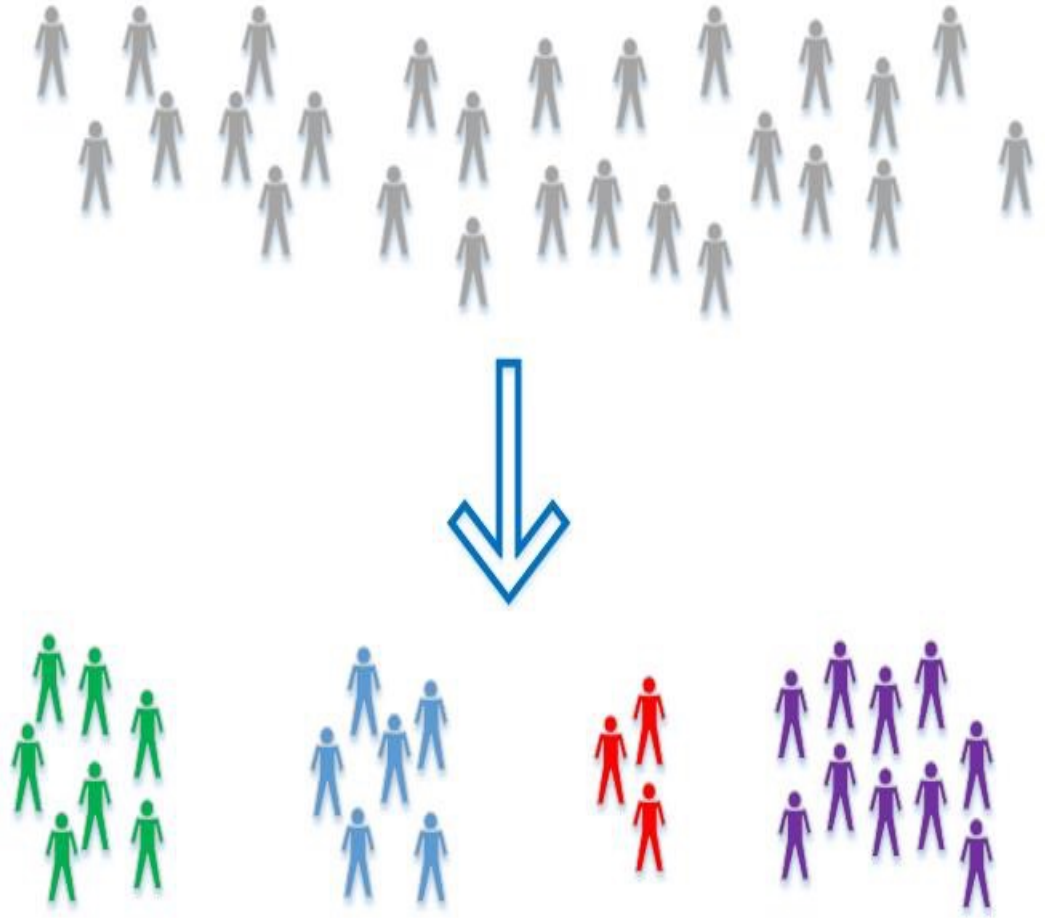


“Over the past decades, computers have broadly automated tasks that programmers could describe with clear rules and algorithms. Modern machine learning techniques now allow us to do the same for tasks where describing the precise rules is much harder.”

Jeff Bezos’ 2017 Letter to Shareholders



UNSUPERVISED MACHINE LEARNING



System looks for naturally occurring patterns, clusters, groupings, or anomalies

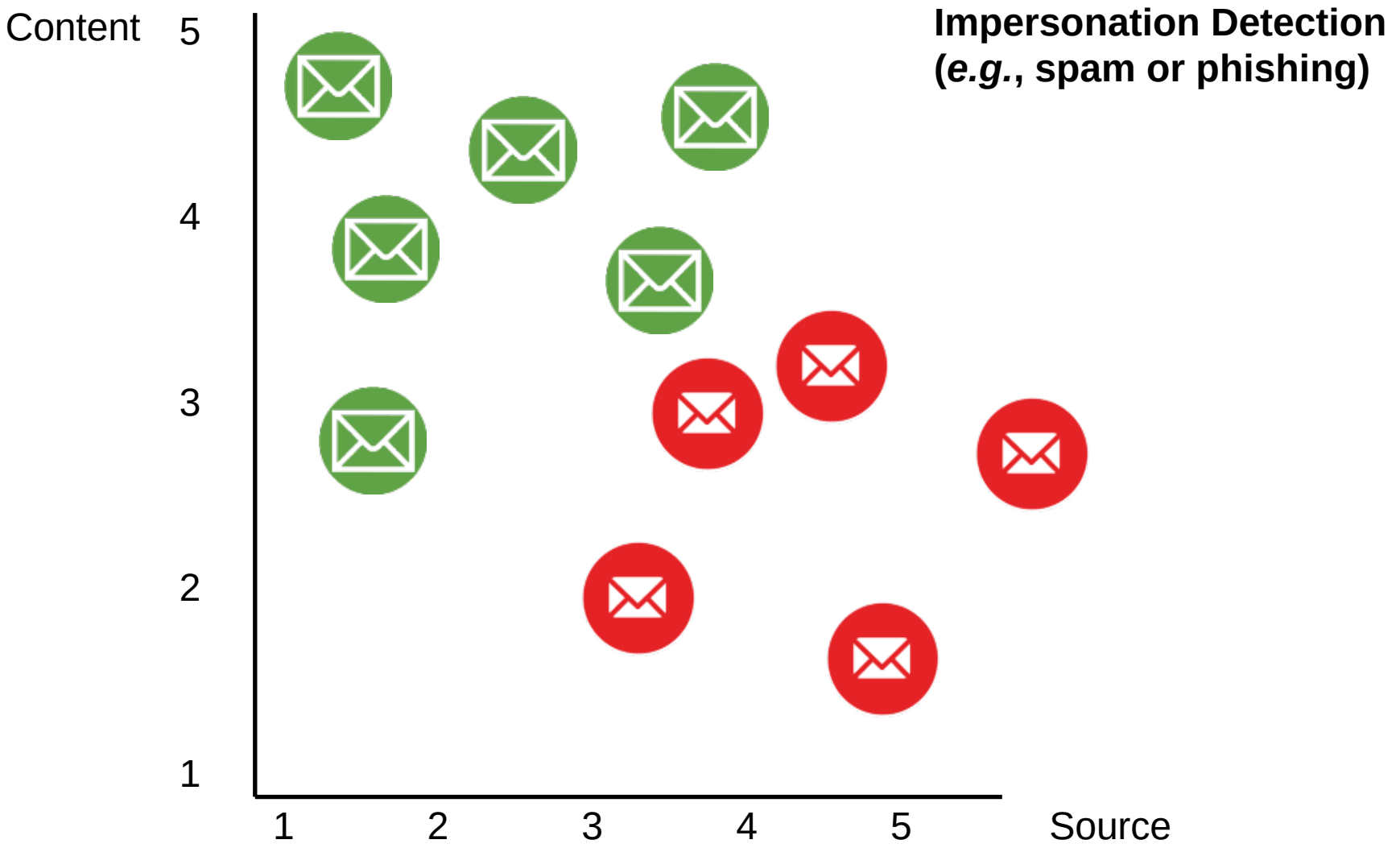
SUPERVISED MACHINE LEARNING



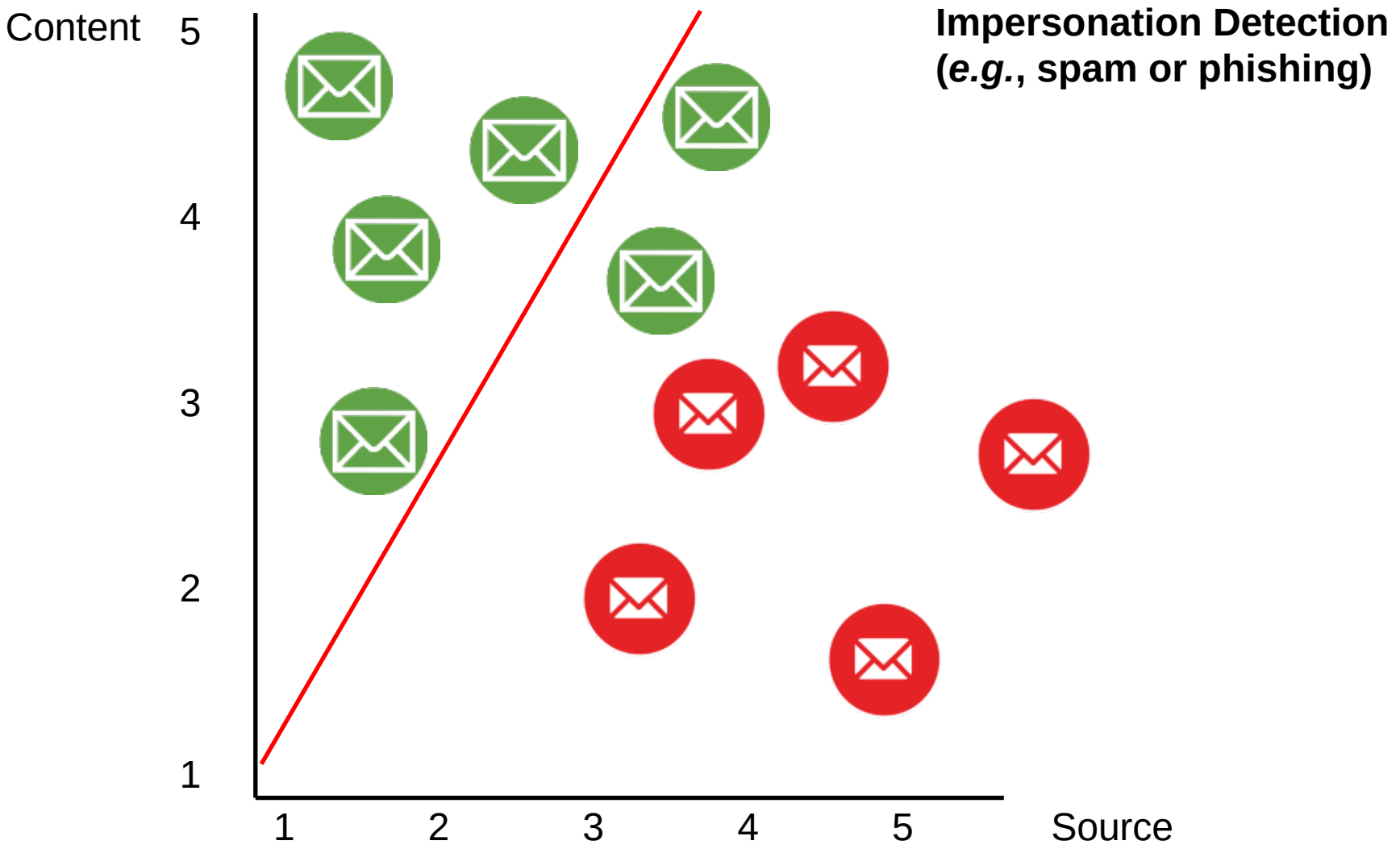


Training a system to distinguish between two or more categories by providing it with labeled examples from which it learns the rules

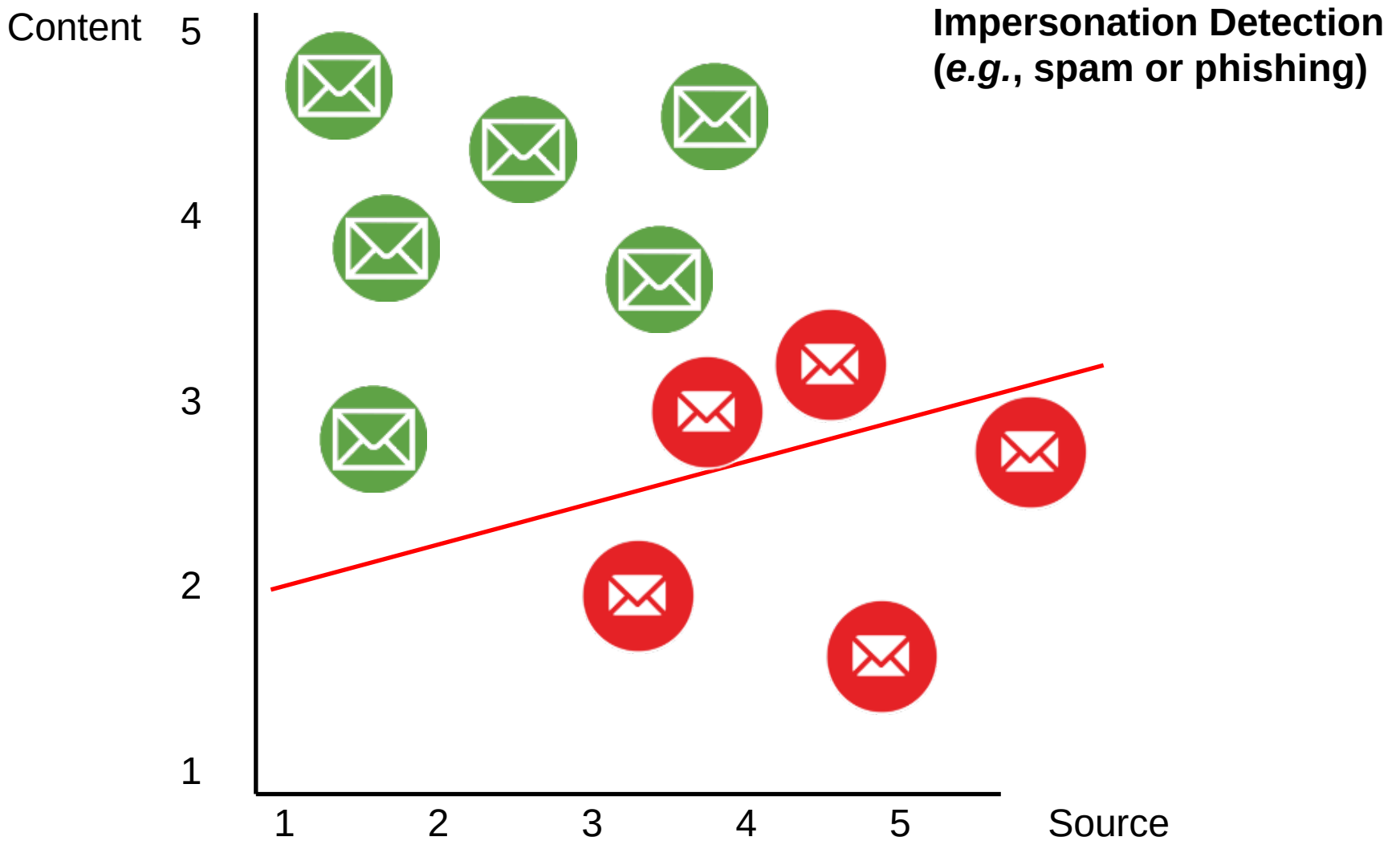
HOW SUPPORT VECTOR MACHINES WORK



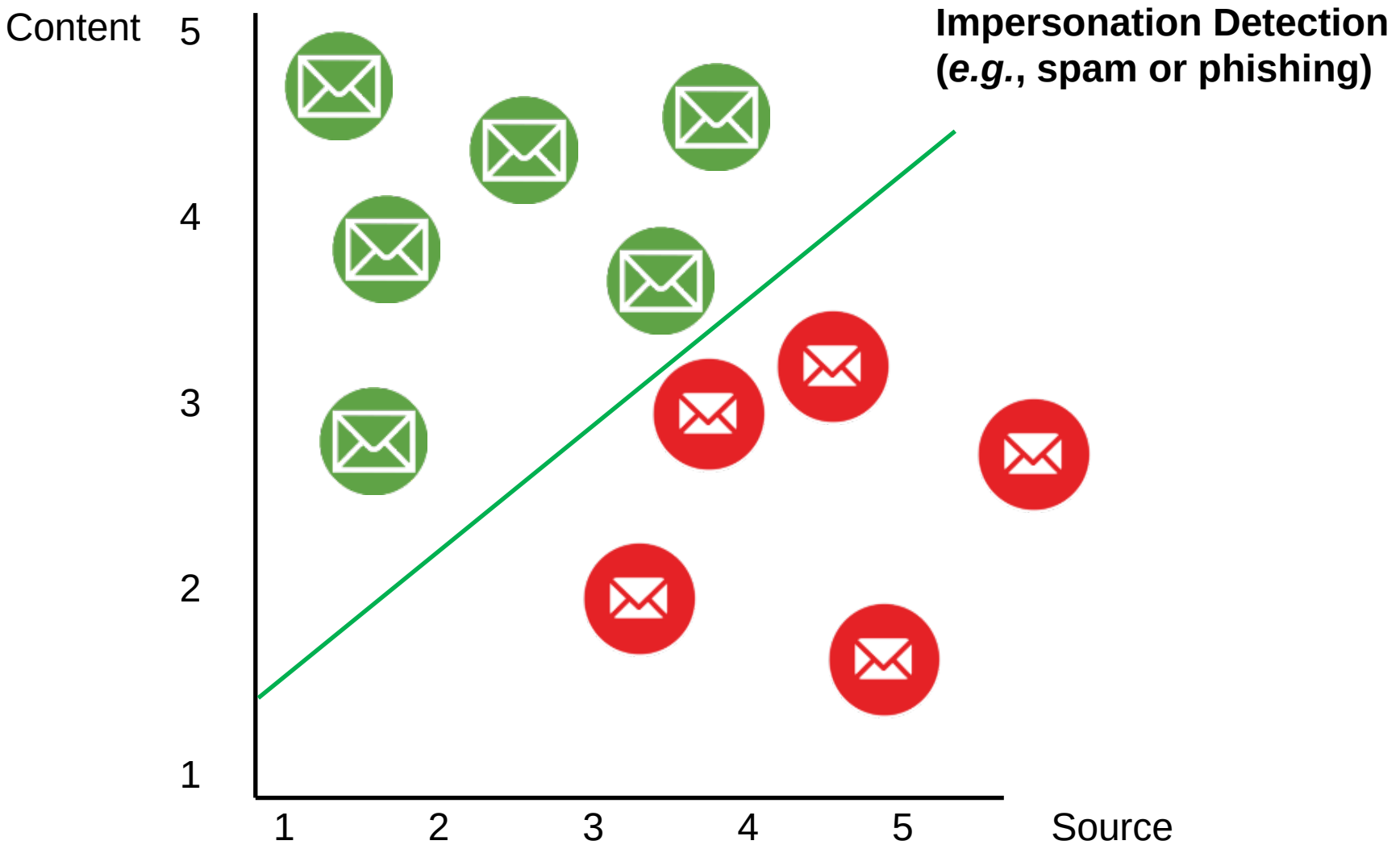
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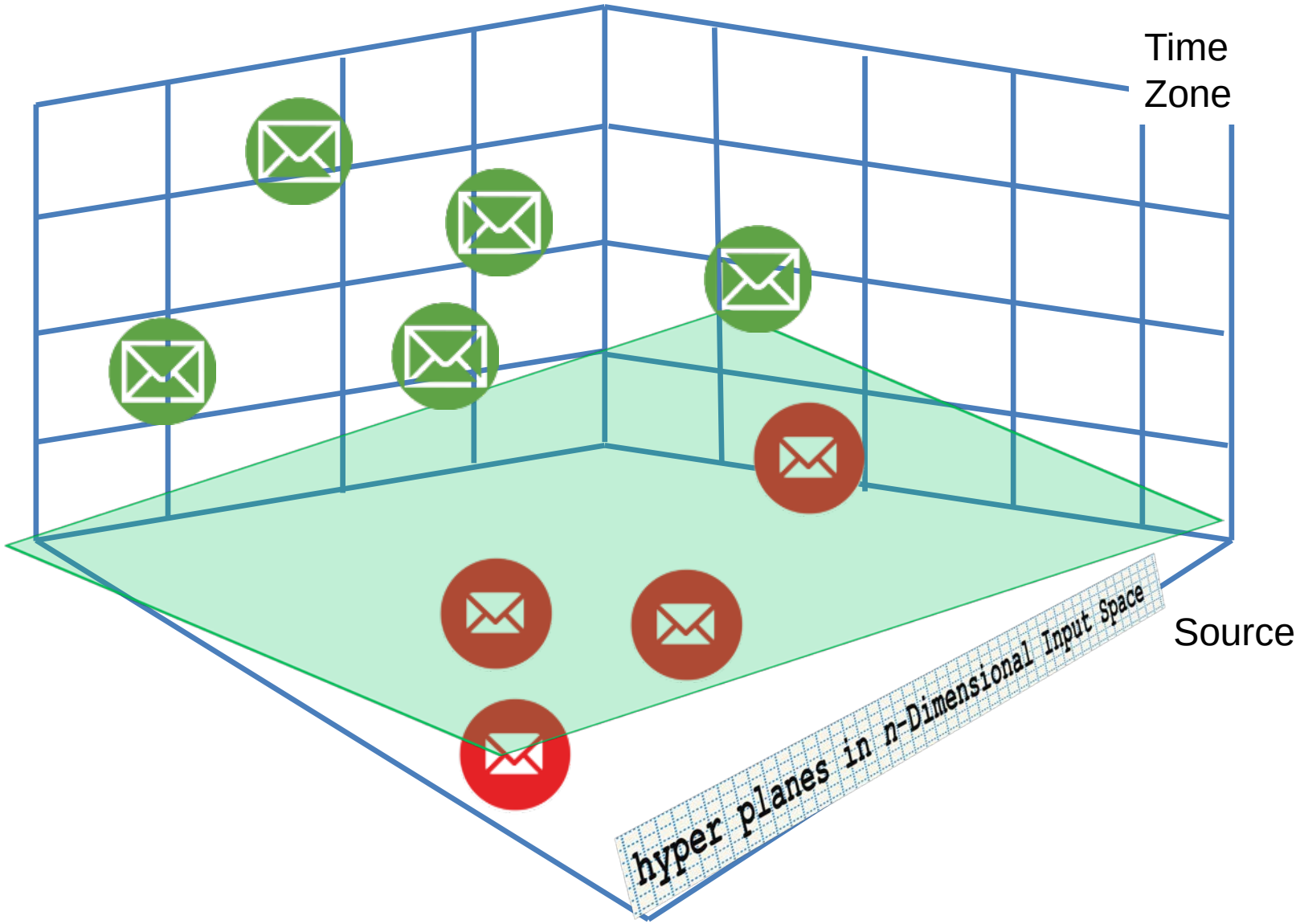
HOW SUPPORT VECTOR MACHINES WORK



NOW IN MULTIPLE DIMENSIONS

Impersonation Detection

- Origin
- Distance
- Device
- Spelling
- Grammar
- Punctuation
- …



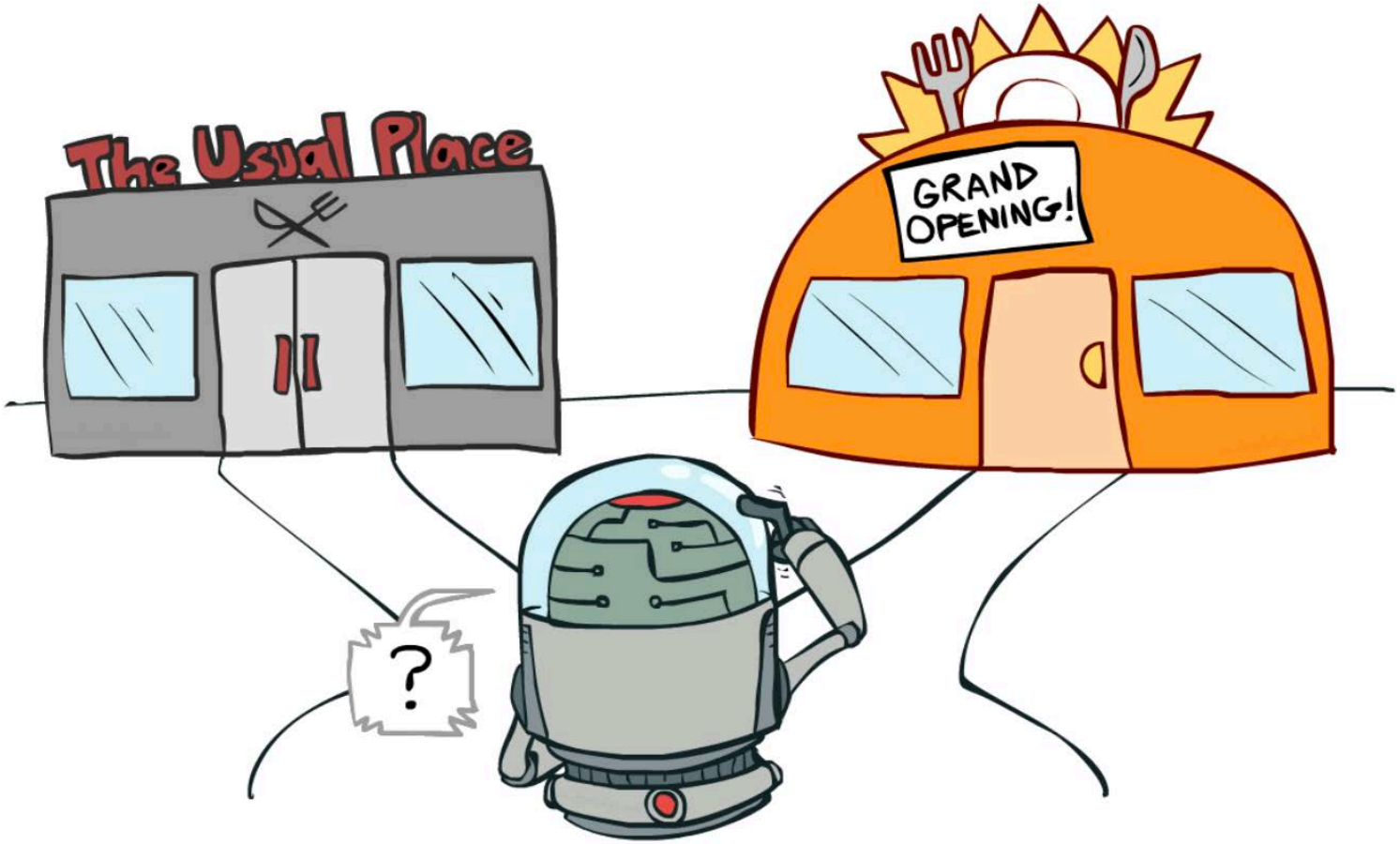
HOW SUPERVISED MACHINE LEARNING WORKS

Supervised machine learning systems infer mathematical functions from old data to help make educated guesses about new data.

JUST MEMORIZE THIS!

$$\begin{cases} \frac{\partial L_1}{\partial w} = 0 & \rightarrow \quad w = \sum_{i=1}^{N} \alpha_i y_i \phi(x_i), \\ \frac{\partial L_1}{\partial b} = 0 & \rightarrow \quad \sum_{i=1}^{N} \alpha_i y_i = 0, \\ \frac{\partial L_1}{\partial \xi_i} = 0 & \rightarrow \quad 0 \leq \alpha_i \leq c, \ i = 1, \ldots, N. \end{cases}$$

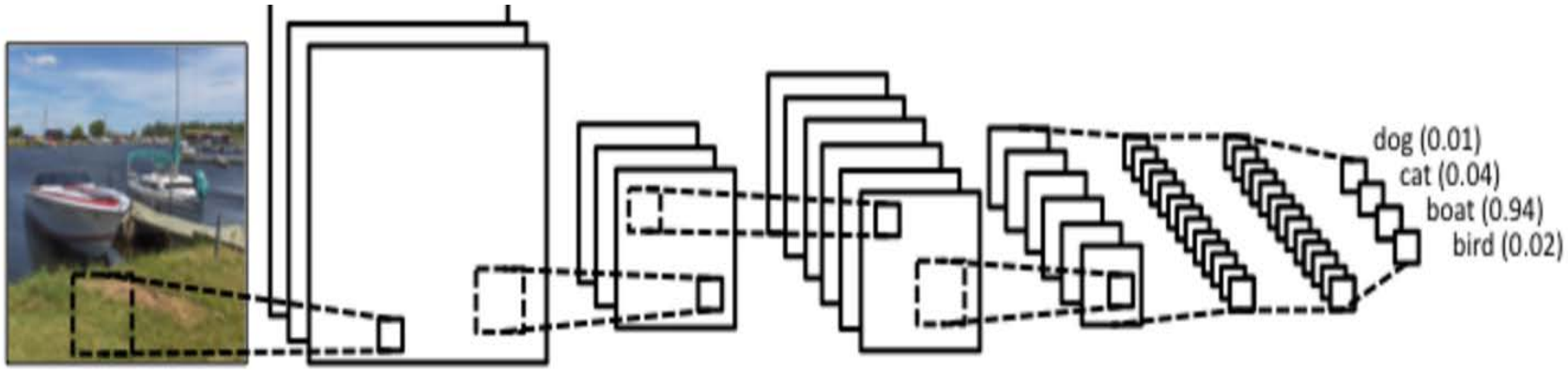
REINFORCEMENT LEARNING



System begins at random but learns goal from reinforcement provided by training

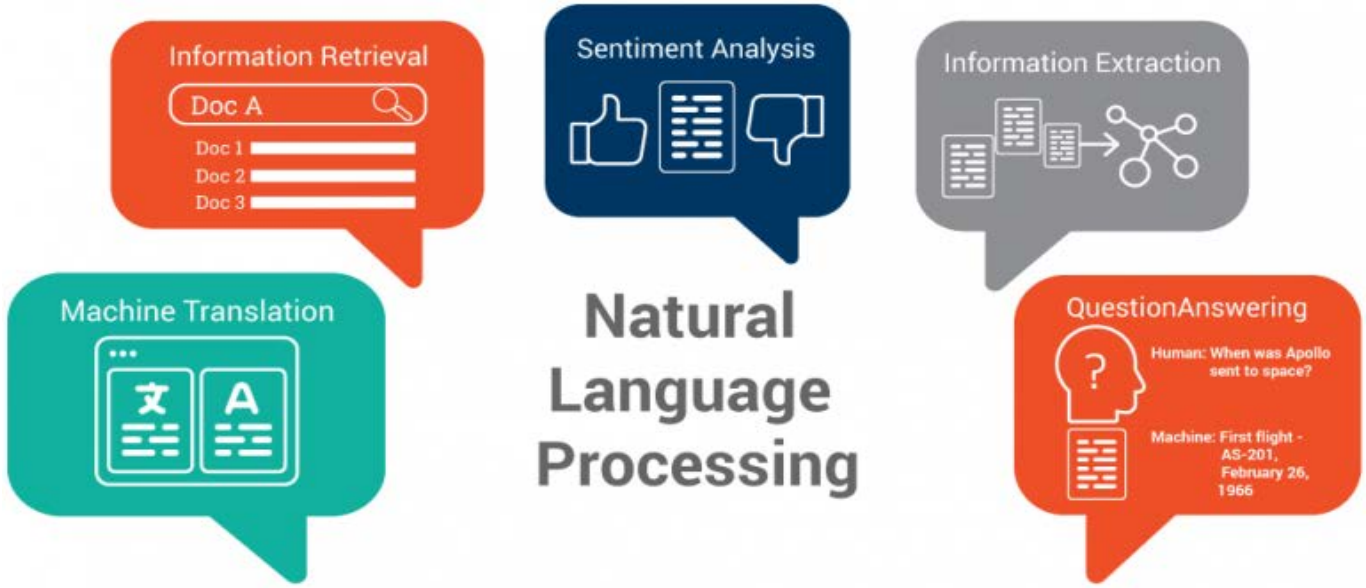
Exploration versus Exploitation

DEEP LEARNING



- Uses multiple layers to transform complex input into mathematical representations
- Information from each layer is combined at the next layer
- Requires massive amounts of labeled training data to work

NATURAL LANGUAGE PROCESSING

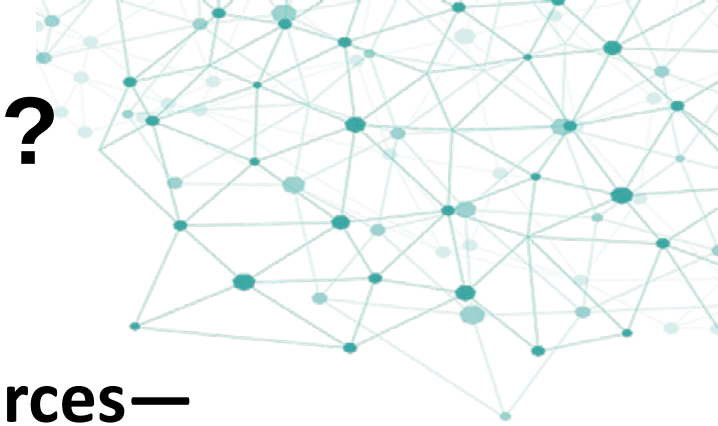


Using a computer to "understand" human language as it is written or spoken, or to create a computer representation of language (including both syntax and semantics)

- **Tokenization** → Splits longer strings into smaller pieces; determines word boundaries
- **Stemming** → Eliminates prefixes and suffixes from words
- **Bag of words** → Looks for co-occurrences of words in a document
- **Stop words** → Removes words that are noise and don't add meaning
- **Tf-idf** → Determines how important a word is to a document by its frequency
- **Disambiguation of content** → Polisemy (*i.e.*, lead vs. lead)
- **Topic modeling** → Statistical models to discover abstract concepts

A Quick Tutorial on Generative AI

WHAT IS “GENERATIVE AI” (“GEN AI”)?



- **A subset of AI that uses training on massive data sources—primarily from the Internet—to generate new content in response to a user prompt. It can converse, replicate specific styles, and excels at creative tasks and synthesizing or summarizing content.**
- **Gen AI falls under the broad categories of machine learning and natural language processing.**
- **It leverages neural networks (*i.e.*, deep learning) to analyze the underlying patterns and structures of data, enabling it to predict what should come next, or to generate fresh and unique content. (This explains why it “hallucinates!”)**

BRIEF HISTORY OF GEN AI: PRECURSORS TO LARGE LANGUAGE MODELS ("LLMS")

- **New or Not?**
  - **Claude Shannon (one of the founders of AI) wrote *Prediction and Entropy of Printed English* in 1951.**
  - **He would erase fragments of text and have humans guess what was erased.**
  - **Language models are statistical models applied to Shannon's prediction task. Historically used for things like data compression, information retrieval, author and spam detection.**
  - **In comes massive computing power + massive data + neural nets, convolutional neural nets, deep learning . . .**

A BRIEF HISTORY OF GEN AI: 2010 – 2022

- In 2014, Generative Adversarial Networks ("GANs") took a huge leap forward in their ability to create authentic-looking content.
- GANs introduce a new way for algorithms to learn: One algorithm (the generative network) creates content, and the other algorithm (the discriminative network) evaluates it against real data in an effort to distinguish them. This approach creates more and more realistic-looking content (and also explains why detection of Gen AI content is so difficult).
- GANs revolutionized image, audio, and video generation.

A BRIEF HISTORY OF GEN AI: 2010 – 2022 (CONT'D)

- In 2017, Google introduced the transformer architecture, a significant breakthrough in processing natural language which no longer required pre-labelled training data and allowed processing to occur in parallel (which is much faster).

- Another major change introduced with GPT-3 was the use of reinforcement learning, in which external (*i.e.*, human) feedback is used to modify and improve the output of the model.

GEN AI IN ACTION: TEXT-TO-TEXT

- **MRG's Prompt:**
  - **"Write a sonnet about eDiscovery in the style of Shakespeare."**

- **ChatGPT's Response: Ode to eDiscovery – A Digital Quest**

Fair denizen of cyberspace, behold!
A wondrous real where knowledge doth reside,
Where bits and bytes, in streams, forever flow,
And secrets deep within the data hide.

In this vast tapestry of virtual threads,
Where truth and fiction dance in pixel's guise,
Lies eDiscovery with keenest treads,
Unveiling tales unseen by human eyes.

Through corridors of code, our search will wend,
With queries cast like spells from ancient lore,
Unraveling the mysteries we intend,
To find the truth that lies behind the door.

Though Shakespeare knew no digital frontier,
His spirit lives in this eDiscovery sphere.

Oh, noble server, guardian of the past,
Reveal the hidden treasures you possess,
From emails sent to messages amassed,
Unveil the truth, unburden our distress.

With algorithms, wielded like a sword,
We navigate the labyrinthian haze,
Each line of code, a clue we can't afford to miss,
As we discover hidden ways.

GEN AI IN ACTION: TEXT-TO-TEXT (CONT'D)

The digital footprints, faint and ethereal,
Betraying secrets tucked away so deep,
Within the bytes, a story empirical,
Emerges from the shadows, no more asleep.

So, let us delve into this brave new world,
Where Shakespeare's verse, with technology's furled.

Through screens and monitors, we find the past,
Deciphering the digital debris,
In search of justice, honest and steadfast,
With eDiscovery our guide to be.

No longer parchment, quill, or ink employed,
But ones and zeros, marks of our advance,
For every whisper, every word deployed,
Leaves traces in the digital expanse.

Yet in this real, where data holds its sway,
We must be wary, mindful of the plight,
For privacy's fragile thread my fray,
As eDiscovery unveils its might.

In Shakespeare's words, we find a truth profound
That eDiscovery wears many a crown.

So let us, with a poet's heart explore,
This brave new world where Shakespeare meets the code,
With eDiscovery, a quest to soar,
And seek the truth where'er it may erode.

In digital domains, where knowledge dwells,
Shall Shakespeare's spirit and technology unite,
As eDiscovery unravels and compels,
To shed light on the darkness, shining bright.

Thus, in this age of virtual pursuit,
We blend the past and future, hand in hand
For in eDiscovery, we find the fruit,
Of Shakespeare's timeless verse, forever grand.

Fair eDiscovery, thy power we commend,
As Shakespeare's legacy and tech transcend.

GEN AI IN ACTION: Text-to-Image

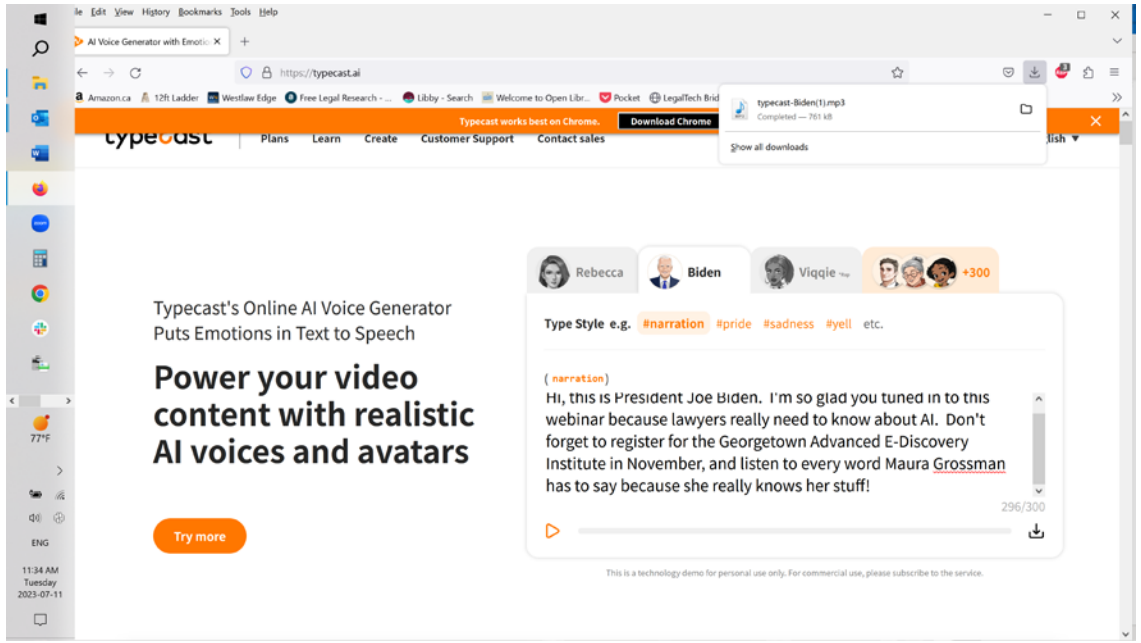
- **MRG's Prompt**:
  - **"Produce a picture of a LGBTQ+ wedding celebration with cats and dogs as guests."**

- **DeepAI's Response (based on Stable Diffusion)**:



GEN AI IN ACTION: Text-to-Speech (“TTS”)

- **MRG’s Prompt:**
  - **President Biden, can you please say hello to these folks and remind them to attend the Georgetown Advanced eDiscovery Institute?”**
- **Typecast’s Response:**



APPLICATIONS OF GEN AI IN LEGAL

- **Gen AI will:**
  - **Enhance delivery of legal services by providing lawyers with tools to increase their productivity**
  - **Enhance access to justice by providing tools to people unable to afford legal services or navigate the legal system**

- **Gen AI will not:**
  - **Replace a lawyer's or judge's reasoning, critical thinking, compassion, empathy, etc.**

- **Gen AI can:**
  - **Analyze and summarize lengthy documents, *e.g.*, complex statutes or regulatory codes; witness transcripts to identify key people, events, or inconsistencies**
  - **Brainstorm ideas**
  - **Help with marketing**
  - **Create outlines and draft documents and presentations**
  - **Conduct research???**
  - **Respond to emails???**



The ChatGPT Associate

- **Draft a complaint for a case to be filed in Illinois State Court, brought by John Doe against Jane Doe for property damage to John Doe's fence, at 1060 W. Addison St., Chicago, arising out of a car accident. Jane Doe was driving at a high speed and lost control of her car driving into John Doe's fence and damaging it.**

- **Draft a rental agreement between landlord, Jane Doe, and renter, John Doe, for a room in the dwelling located at 1060 W. Addison St., Chicago, for a period of one year, beginning on May 1, 2023, and ending on April 30, 2024, for $1,000 per month.**

RISKS OF GEN AI IN LEGAL

- Gen AI does not respect confidentiality or privacy;
  anything you enter may be used for training or other purposes

- Gen AI does not guarantee the accuracy of its output
  - It sounds very confident and compelling
  - But, . . . it hallucinates
  - It reinforces stereotypes (as you saw!)
  - It is predicting things based on Internet content; your mileage may vary
  - It can be biased, toxic, and defamatory

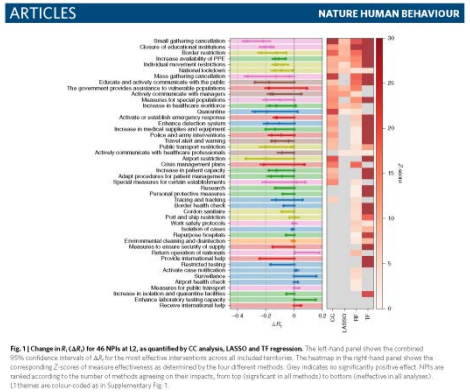
- Gen AI is not secure and is subject to jailbreaking and other adversarial attacks

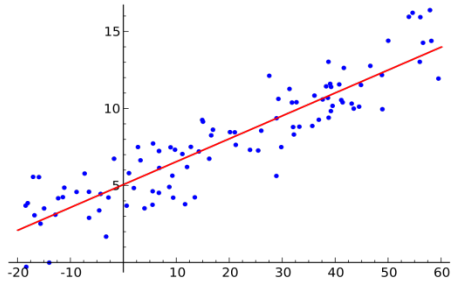
- Gen AI content is likely not subject to copyright protection and may infringe on others' IP

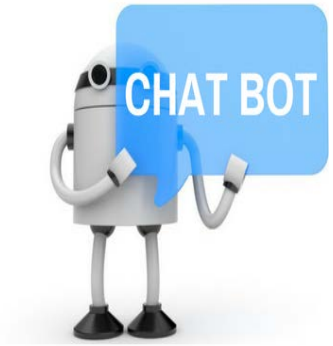
What AI Can Do

WHAT KINDS OF PROBLEMS CAN AI SOLVE?









Classification or Prediction

Relevant / Not Relevant

Ranking

Most to least likely to be relevant

Regression

Numerical estimate of probability of relevance

Generating, Conversing or Translating

“It’s relevant!”

AI TODAY: THE SKY'S THE LIMIT

- Health Care
  - Detecting cancer in MRIs
  - Diagnosing illness or prescribing treatment
- Education
  - Evaluating teacher performance
- Employment
  - Evaluating job applicants for interviews
- Transportation
  - Autonomous vehicles
  - Drones
  - Logistics
- Finance
  - Credit forecasting
  - Mortgage underwriting
- Law Enforcement
  - Predictive policing and facial recognition
  - Bail evaluation and sentencing
- Government
  - Benefits determination
  - Fraud detection
  - Autonomous weapons

AI IN THE LAW

- Technology-assisted review (“TAR”) and other analytics in eDiscovery
- M&A due diligence / contract analysis and review
- Public disclosure analytics
- Legal research / summarization / drafting of memos and initial pleadings
- Analysis of briefs for missing citations
- Predictions about opposing counsel and courts
- Litigation outcome forecasting for financing purposes
- Jury pool evaluation
- Analysis of claims and automated forms completion
- Billing
- …

Some Issues Implicated by AI

AI RAISES A SLEW OF LEGAL AND OTHER ISSUES

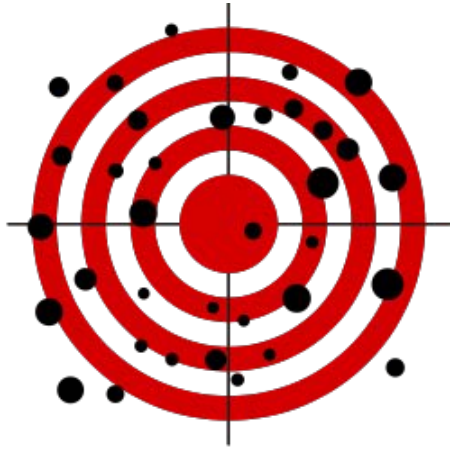


- Bias
  - Data
  - Algorithm
  - Human
    - ✓Algorithm Aversion
    - ✓Automation Bias
    - ✓Confirmation Bias
- Lack of Robust Testing
  - Validity → Does in measure what it is supposed to measure?
  - Reliability → Does it measure it consistently?
  - False Positives vs. False Negatives
- "Function Creep"

VALIDITY VERSUS RELIABILITY



Unreliable & Unvalid



Unreliable, But Valid



Reliable, Not Valid



Both Reliable & Valid

OTHER LEGAL AND ETHICAL ISSUES



- Claims of trade secret leading to lack of due process
- Lack of transparency and explainability
- Lack of resilience to intentional and unintentional efforts to cause the system to perform other than as designed
- Lack of competence
- Lack of data privacy / security
- Lack of accountability and regulation
- . . .



OPPORTUNITY OR DISPLACEMENT?

"Using technology to solve specific real problems using the advantages that computers have with the special abilities that we as human beings share."

Questions? Thank You!

- **Contact Information**

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